

$$\begin{aligned} \text{Case 7: } y=6 \quad n=10x+6 \quad \therefore n^4 &= (10x+6)^4 = (100x^2+120x+36)^2 \\ &= 10000x^4 + 14400x^3 + 1296 + 24000x^2 + 8640x + 7200x^2 \\ &= 10(1000x^4 + 1440x^3 + 129 + 2400x^2 + 864x + 720x^2) + 6 \end{aligned}$$

It's easy to see that n^4 ends with 6 because $10(1000x^4 + 1440x^3 + 129 + 2400x^2 + 864x + 720x^2)$ ends with a 0, since it's a multiple of 10. $\square$

$$\begin{aligned} \text{Case 8: } y=7 \quad n=10x+7 \quad \therefore n^4 &= (10x+7)^4 = (100x^2+140x+49)^2 \\ &= 10000x^4 + 19600x^3 + 2401 + 28000x^2 + 13720x + 9800x^2 \\ &= 10(1000x^4 + 1960x^3 + 240 + 2800x^2 + 1372x + 980x^2) + 1 \end{aligned}$$

$\hookrightarrow$ it's a multiple of 10, $\therefore$ it ends with 0.

$\therefore n^4$ ends with 1 $\square$

$$\begin{aligned} \text{Case 9: } y=8 \quad n=10x+8 \quad \therefore n^4 &= (10x+8)^4 = (100x^2+160x+64)^2 \\ &= 10000x^4 + 25600x^3 + 4096 + 32000x^2 + 20480x + 12800x^2 \\ &= 10(1000x^4 + 2560x^3 + 409 + 3200x^2 + 2048x + 1280x^2) + 6 \end{aligned}$$

$\hookrightarrow$ it's a multiple of 10, $\therefore$ it ends with 0.

$\therefore n^4$ ends with 6 $\square$

$$\begin{aligned} \text{Case 10: } y=9 \quad n=10x+9 \quad \therefore n^4 &= (10x+9)^4 = (100x^2+180x+81)^2 \\ &= 10000x^4 + ~~12100~~ 32400x^3 + 6561 + 36000x^2 + 29160x + 16200x^2 \\ &= 10(1000x^4 + 3240x^3 + 656 + 3600x^2 + 2916x + 1620x^2) + 1 \end{aligned}$$

$\hookrightarrow$ it ends with 0, since it's a multiple of 10. $\therefore n^4$ ends with 1 $\square$

~~Conclusion:~~ Conclusion: n^4 ends with 0, 1, 5 or 6 $\square$